

# SEL (Schweitzer Engineering Laboratories) Firmware Attack Surface & Detection Brief

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**Scope:** 6 product lines / 14,255 unique hashes across 4 architecture classes (SEL-3355-2 industrial PC: 1,127; SEL-BaRT backup-recovery: 12,117; SEL Compass setup tool: 969; SEL-3300/3355 non-2: 16; Virtual Port Service 5828: 22). Findings combine CVE/PSIRT research with direct examination of the extracted Windows-side firmware and engineering-tool binaries (services, default configs, embedded resources). The relay (SEL-300/400/700) and RTAC (SEL-3530/3555) firmware is NOT in this extraction — those rows are research-only and explicitly marked.

**Purdue layer mapping:** Industrial PCs and engineering software (Groups A/B) live at **Purdue L3 (Site Operations)** as the substation EWS / HMI host tier; protective relays and RTAC comms processors (Groups C/D) sit at **Purdue L1 (Basic Controllers)**. See [purdue-l3-site-operations-brief.md](#) and [purdue-l1-basic-controllers-brief.md](#) for cross-vendor views.

## Architecture grouping (drives the threat model, not the SKU)

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| Class                       | Purdue layer       | Products                                                               | Stack                                                                                                      | Catalog depth |
|-----------------------------|--------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------|
| A. SEL Industrial PCs       | L3 Site Operations | SEL-3355-2, SEL-3355, SEL-3300                                         | Windows 10 IoT LTSC / Win Server, ruggedized substation HMI/SCADA host, runs SEL acSElerator + RTU clients | 1,143 hashes  |
| B. SEL engineering software | L3 Site Operations | acSElerator QuickSet, SEL-BaRT, SEL Compass, Virtual Port Service 5828 | Windows .NET / native, serial-over-TCP virtualization, relay config & firmware-push tooling                | 13,130 hashes |

| Class                                               | Purdue layer         | Products                                       | Stack                                                                                         | Catalog depth |
|-----------------------------------------------------|----------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------|
| <b>C. Protective relays (research only)</b>         | L1 Basic Controllers | SEL-300/351/387/400/421/451/487/700/751 series | Bare-metal embedded (proprietary RTOS), Fast Message / SEL ASCII / IEC 61850 GOOSE+MMS / DNP3 | not extracted |
| <b>D. Communications processors (research only)</b> | L1 Basic Controllers | SEL-3530 / SEL-3555 RTAC, SEL-2730M switch     | Embedded Linux (RTAC OS), serves IEC 61850 / DNP3 / Modbus / IEC 60870-5-104 concentration    | not extracted |

## Group A — SEL Industrial PCs (3355/3300 substation HMI hosts) — Purdue L3 (Site Operations)

**Direct attack surface (verified via PE imports and embedded `.inf` /service configs in the extracted 3355-2 catalog):**

SMB • RDP • WinRM • SEL-5045 ICS Studio • acSELeator runtime • SEL Virtual-Port-Service (TCP serial bridge) • USB host

The 3355-2 ships as a ruggedized Windows IoT box. Embedded driver packages in the extracted hashes include the SEL serial-port virtualization stack (Virtual-Port-Service 5828) and acSELeator helper services — both run as `LocalSystem` and listen on configurable TCP ports for serial-tunneling. Default Windows posture (SMB enabled, RDP often enabled for substation remote-ops) carries.

### Confirmed advisories / CVEs:

| Advisory                            | Product                 | Vector                                                                                                      | Status on the firmware you have               |
|-------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <a href="#">CISA ICSA-23-131-08</a> | SEL-5030 QuickSet, SEL- | Bundled advisory covering CWE-276 insecure default permissions, CWE-22 path traversal, and CWE-269 improper | acSELeator binaries present in BaRT catalog — |

| Advisory                                      | Product                                           | Vector                                                                                                                                                  | Status on the firmware you have                          |
|-----------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|                                               | 5033 RTAC, SEL-5045 ICS Studio                    | privilege management in the acSELerator product family                                                                                                  | verify versions against the advisory's fixed-build table |
| <a href="#">SEL Security Advisories index</a> | acSELerator QuickSet (SEL-5030)                   | XML external entity (XXE) handling on relay project ( .ac1x ) import → local file disclosure / SSRF; addressed in vendor-published 2023 advisory bundle | Triage operator workstations that open relay configs     |
| <a href="#">CISA ICSA-23-194-02</a>           | SEL Real-Time Automation Controller (RTAC) family | Multiple weaknesses in RTAC firmware and management interfaces                                                                                          | Co-installed RTAC-management binaries staged via Compass |

**Top attack vector (MITRE ATT&CK ICS):** [T0817 Drive-by Compromise](#) of the engineering workstation followed by [T0843 Program Download](#) — once acSELerator is compromised, the operator's own session pushes tampered logic to downstream relays. INL's CRASHOVERRIDE post-mortem and Dragos's PIPEDREAM analysis both flag engineering workstations as the highest-value foothold in substation environments.

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## Group B — SEL engineering software (acSELerator / BaRT / Compass / VPS) — Purdue L3 (Site Operations)

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**Direct attack surface (from the 12,117 BaRT hashes + 969 Compass hashes + 22 VPS binaries):**

SEL-BaRT is the relay backup/restore tool — it speaks SEL Fast Message and SEL ASCII over serial or TCP-virtualized serial to read/write **relay settings groups and firmware**. SEL Compass is the launcher/updater that authenticates to SEL's update servers and stages firmware payloads to disk. Virtual Port Service 5828 exposes a TCP listener (default 23xxx range) that bridges to local COM ports — any code that can talk to that listener can drive an attached relay as if it had physical serial access.

**Confirmed advisories / CVEs:**

| Advisory                                      | Product                                | Vector                                                                                                           | Status on the firmware you have                                                              |
|-----------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <a href="#">CISA ICSA-23-131-08</a>           | acSElerator QuickSet (SEL-5030)        | DLL search-order / writable-install-path class enabling local privilege escalation to SYSTEM via service restart | Audit DLLs in the BaRT/Compass install trees you extracted; flag any unsigned or non-SEL DLL |
| <a href="#">SEL Security Advisories index</a> | SEL-5037 Grid Configurator             | Cleartext / weakly-protected storage of project credentials in configuration DB                                  | Grid Configurator not in this dataset, but commonly co-installed                             |
| <a href="#">CISA ICSA-23-131-08</a>           | acSElerator Team (SEL-5045) ICS Studio | Improper authentication / authorization on local IPC + management endpoints                                      | Co-installed with ICS Studio binaries in this catalog                                        |
| <a href="#">SEL Security Advisories index</a> | acSElerator Architect (SEL-5032)       | Authorization / project-import handling weaknesses (2024 vendor advisory cycle)                                  | Architect commonly bundled in Compass-managed installs                                       |

**Top attack vector:** [T0822 External Remote Services](#) → [T0857 System Firmware](#). The BaRT-style backup/restore flow is the canonical relay-firmware-tamper path; ICEFALL-class trust-bus weaknesses in OT engineering tools were the centerpiece of Forescout's 2022 disclosure series.

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## Group C — Protective relays (research only, firmware NOT extracted) — Purdue L1 (Basic Controllers)

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SEL relays (300/351/387/400/421/451/487/700/751 series) are bare-metal embedded with proprietary code-signed firmware. They expose:

- **SEL Fast Message** and **SEL ASCII** on serial / serial-over-TCP — the native control plane (read settings, write settings, trip/close breakers, reset targets).
- **IEC 61850 MMS** (TCP/102) and **GOOSE** (Ethertype 0x88B8) on Ethernet-enabled variants.
- **DNP3** (TCP/20000) and **IEC 60870-5-104** (TCP/2404) where licensed.

**Attack vectors (no extraction-confirmed CVEs — assume CVE history is thin due to closed source):**

- **GOOSE injection** — INL/DOE red-team work (2019-2023) repeatedly demonstrated unauthenticated GOOSE on substation LANs causes unintended breaker operation. Mitigation is GOOSE MAC (IEC 62351-6), rarely deployed.
- **Settings-group switch via Fast Message** — operator-equivalent commands; if attacker reaches the relay's serial or virtualized-serial bridge they can swap protection settings.

- **Manufacture-time / default level passwords** — SEL relays historically shipped with well-known default level-1/2/C passwords ( `OTTER` , `TAIL` , `CLARKE` ); INL pen-tests still find these unchanged in the field.

**Top vector:** [T0855 Unauthorized Command Message](#) via GOOSE or Fast Message from a compromised engineering workstation. CRASHOVERRIDE used precisely this pattern against Ukrenerg (different relay vendor, identical class of attack).

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## Group D — RTAC communications processors (research only) — Purdue L1 (Basic Controllers)

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SEL-3530 / SEL-3555 RTAC runs an embedded Linux ("RTAC OS"). It concentrates protocols (DNP3 master/slave, IEC 61850 client, Modbus, IEC 60870-5-104) and runs IEC 61131-3 logic. Public CVE coverage in this family is sparse; the [CISA ICSA-23-194-02](#) RTAC advisory bundle is the public anchor. Attack-surface assumption: HTTPS web UI on TCP/443, SSH on TCP/22 (factory-disabled but operator-enabled in the field), DNP3 outstation, and the IEC 61850 client/server stack.

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## Logging matrix (highest priority)

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Top 8 — ordered by detection value per ingest dollar:

| # | Source                                        | Event                                                                                                                                                              | What it catches                                     | MITRE ATT&CK ICS                             |
|---|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| 1 | <b>Sysmon on SEL engineering workstations</b> | EventID 1 ProcessCreate where ParentImage= <code>AcSEleratorQuickSet.exe</code> / <code>BaRT.exe</code> / <code>Compass.exe</code> AND child not in SEL allow-list | acSElerator post-exploitation / firmware-push abuse | <a href="#">T0843 Program Download</a>       |
| 2 | Sysmon                                        | EventID 7 ImageLoad on <code>AcSElerator*.exe</code> / <code>BaRT.exe</code> loading DLLs from user-writable paths                                                 | DLL search-order hijack LPE class                   | T1574.001                                    |
| 3 | Sysmon                                        | EventID 11 FileCreate of <code>*.aclx</code> / <code>*.RDB</code> / <code>*.crd</code> from non-engineer accounts                                                  | Tampered relay-settings project staging             | <a href="#">T0873 Project File Infection</a> |
| 4 | <b>Network IDS (Zeek/Suricata)</b>            | TCP/102 MMS <code>write</code> services from non-engineering hosts                                                                                                 | IEC 61850 settings tamper                           | <a href="#">T0836 Modify Parameter</a>       |

| # | Source                                                       | Event                                                                                                | What it catches                        | MITRE ATT&CK ICS                                   |
|---|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------|
| 5 | Network IDS                                                  | GOOSE frames (Ethertype 0x88B8) sourced from non-relay MACs, or stNum jumps > expected               | GOOSE spoofing precursor               | <a href="#">T0855 Unauthorized Command Message</a> |
| 6 | Network IDS                                                  | DNP3 function codes 5 (DirectOperate) / 6 (DirectOperateNoAck) crossing engineering→OT boundary      | Breaker trip/close abuse               | <a href="#">T0831 Manipulation of Control</a>      |
| 7 | Network IDS                                                  | Connections to <b>Virtual Port Service 5828</b> TCP listener from non-loopback / non-engineering IPs | Lateral relay access via serial tunnel | <a href="#">T0886 Remote Services</a>              |
| 8 | <b>SEL relay SER / Sequential Events Recorder</b> via syslog | Settings-group change, password-attempt failures, breaker operation outside scheduled windows        | Live relay tamper detection            | <a href="#">T0858 Change Operating Mode</a>        |

## Secondary:

- **Firewall:** any egress from substation engineering VLAN to non-RFC1918 (engineering workstations should never originate outbound to internet — SEL Compass updates can be staged via an internal mirror).
- **SEL relay audit logs:** every SEL-400/700-series relay has a built-in SER buffer and Audit log accessible via Fast Message `HIS` / `MET`. Pull on schedule and diff for unexpected setting-group changes, level-2/C password failures, and `RST TARGET` events.
- **Compass update channel:** monitor outbound TLS to `updates.selinc.com` / `compass.selinc.com` — only the Compass-host should originate these; any other host doing so is a staging artifact.

## Power-anomaly signals (the OT-native detection):

- Unexpected breaker open/close events in relay SER buffer outside maintenance windows
- Setting-group switch (relay reports `SG=N` change) with no concurrent operator-ticket trail
- IEEE C37.118 synchrophasor stream gaps / quality-flag changes on PMU-capable relays — control-loop tamper precursor

## Specific zero-day-ish concerns for your dataset

1. **acSELERator DLL-hijack / writable-install-path class (covered by ICSA-23-131-08 and the SEL 2023 advisory bundle) on a substation workstation is the operator-equivalent compromise.** The 1,127 SEL-3355-2 hashes plus 969 Compass hashes are exactly the

engineering-workstation profile. Diff the DLL imports in your extracted catalog against SEL's signed-binary manifest; any unsigned or non-SEL DLL in the install tree is a planted hijack candidate.

2. **SEL-BaRT (12,117 hashes — the largest single product in the extraction) is the firmware-push tool.** Its trust relationship with downstream relays is unauthenticated at the Fast Message layer. Any compromise of BaRT.exe or its config DB = silent relay-firmware tamper across the substation. Treat BaRT-host binaries as crown-jewel and integrity-monitor them at file-hash level.
3. **Virtual Port Service 5828 (22 hashes) — small but high-blast-radius.** VPS exposes a local TCP listener that bridges to physical COM ports. Verify in the extracted binaries (a) what interface it binds to (loopback vs 0.0.0.0), (b) whether it enforces any auth, (c) what port range it uses by default. If it binds to 0.0.0.0 with no auth, it is a remote serial-console-equivalent on the management VLAN.
4. **Public CVE coverage against SEL relay firmware itself is thin,** despite INL/DOE red-team reports describing exploitable conditions. Closed-source + light public scrutiny ≠ no bugs. Plan extraction of SEL-3530/3555 RTAC firmware next — it is the embedded-Linux concentrator and the easiest pivot into the relay LAN.

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## Sources

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- [CISA ICS Advisory ICSA-23-131-08 — SEL acSElerator multiple products](#)
- [CISA ICS Advisory ICSA-23-194-02 — SEL Real-Time Automation Controller \(RTAC\)](#)
- [SEL Security Advisories index](#)
- [Forescout — OT:ICEFALL research \(engineering-tool trust-bus class\)](#)
- [Dragos — PIPEDREAM/CHERNOVITE analysis \(engineering-workstation pivot\)](#)
- [INL — Cyber-Informed Engineering and substation red-team findings](#)
- [MITRE ATT&CK for ICS — Initial Access \(TA0108\)](#)
- [MITRE ATT&CK for ICS — Impair Process Control \(TA0106\)](#)
- [MITRE ATT&CK for ICS — T0857 System Firmware](#)
- [MITRE ATT&CK for ICS — T0855 Unauthorized Command Message](#)
- [MITRE ATT&CK for ICS — T0836 Modify Parameter](#)
- [MITRE ATT&CK for ICS — T0843 Program Download](#)
- [CISA/NSA/FBI/ASD — Best Practices for Event Logging and Threat Detection \(2024\)](#)
- [Claroty Team82 — research blog \(ICS engineering-tool advisories\)](#)